

Mini Project 02: Fully-Differential Folded Cascode OTA

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1 Part 1: g_m/I_D Design Charts

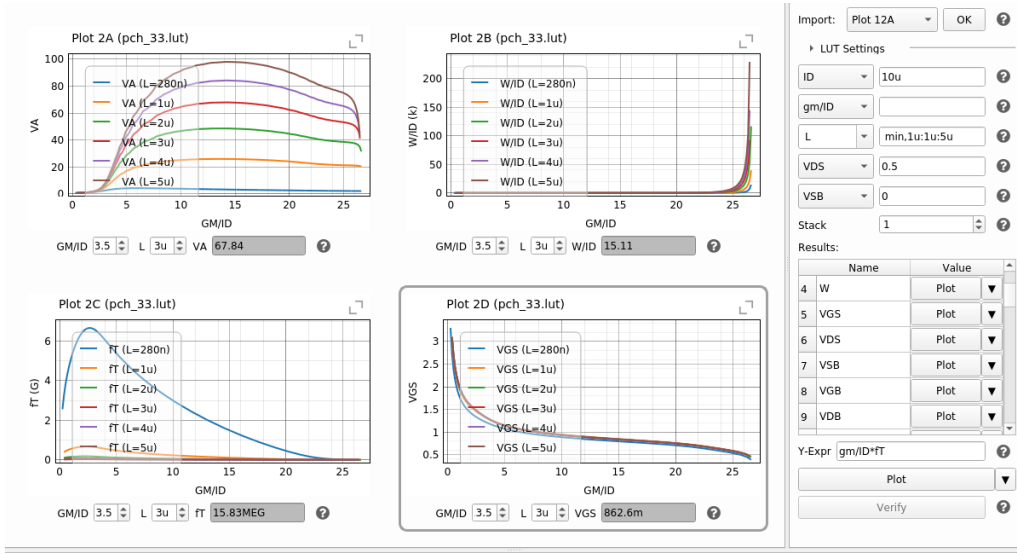


Figure 1: PMOS Design Charts vs. g_m/I_D (V_A , W/I_D , f_T , V_{GS})

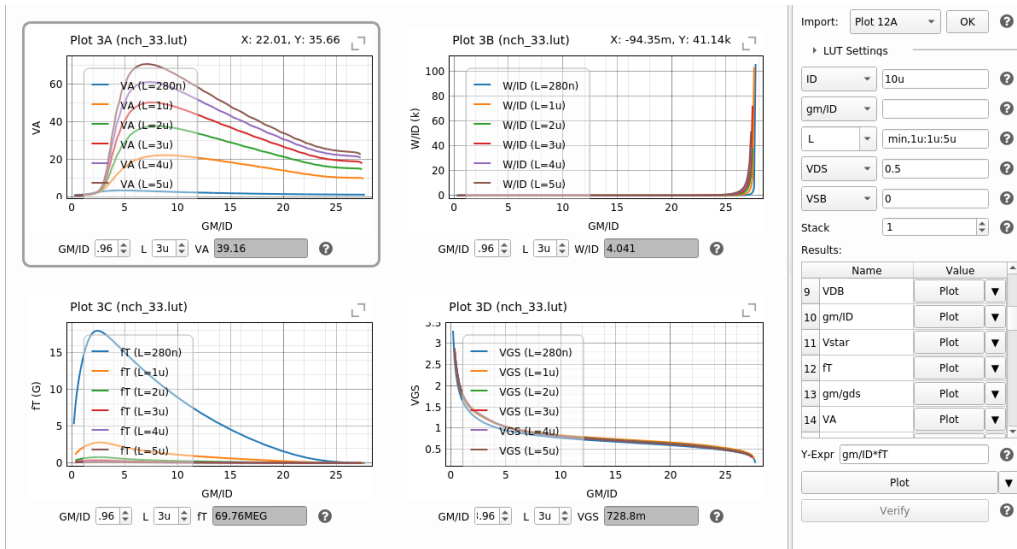


Figure 2: NMOS Design Charts vs. g_m/I_D (V_A , W/I_D , f_T , V_{GS})

2 Sizing and Hand Analysis

2.1 Target Specifications

The design is implemented using the GF180MCU technology node with a 2.5 V supply. The target specifications for the fully-differential folded cascode OTA are summarized in Table 1.

Table 1: GF180MCU Target Specifications

Parameter	Specification
Technology	GF180MCU
Supply Voltage (V_{DD})	2.5 V
Closed Loop Gain	2 V/V
Phase Margin	$\geq 70^\circ$
CM Input Range - Low	≤ 0 V
CM Input Range - High	≥ 1 V
Differential Output Swing	1.2 V _{pk-to-pk}
Load Capacitance (C_L)	500 fF
DC Loop Gain	60 dB
CL Settling Time (1% error)	100 ns

2.2 Hand Analysis for Input Transconductance (g_{m1})

To achieve the 100 ns settling time for a 1% error, the required closed-loop bandwidth (ω_{CL}) is first calculated assuming a single-pole dominant system:

$$e^{-t_s/\tau} = 0.01 \implies \frac{100 \text{ ns}}{\tau} \approx 4.6 \implies \tau \approx 21.7 \text{ ns} \quad (1)$$

$$\omega_{CL} = \frac{1}{\tau} \approx 46 \text{ Mrad/s} \quad (2)$$

The closed-loop bandwidth is related to the input transconductance (g_{m1}), the feedback factor (β), and the total effective load capacitance ($C_{L,eff}$) by:

$$\omega_{CL} = \beta \cdot \omega_{GBW} = \beta \left(\frac{g_{m1}}{C_{L,eff}} \right) \quad (3)$$

Accounting for the ideal feedback factor ($\beta = 1/3$) and the combined loading of the 500 fF load capacitor along with the feedback network, the baseline hand calculation dictates a required transconductance of:

$$g_{m1(calc)} \approx 160 \mu\text{S} \quad (4)$$

However, to introduce a safety margin that compensates for parasitic capacitances at the input and folding nodes, the target transconductance was increased to $180 \mu\text{S}$.

To maximize power efficiency while achieving this chosen target, the input pair is biased in moderate-to-weak inversion. Selecting a branch bias current of $I_D = 10 \mu\text{A}$ yields the design parameter:

$$\frac{g_m}{I_D} = \frac{180 \mu\text{S}}{10 \mu\text{A}} = 18 \text{ V}^{-1} \quad (5)$$

We increased g_m to $180 \mu\text{S}$ to give some margin in GBW and help meet the 100 ns settling-time requirement.

2.3 Analog Designer's Toolbox (ADT) Extraction

The following subsections detail the transistor sizing utilizing the g_m/I_D methodology and the ADT simulation charts.

2.3.1 Input Pair (PMOS)

For the PMOS input pair, V_{DS} is set to 0.5 V to ensure deep saturation while leaving sufficient voltage headroom for the tail and cascode branches. Based on the target $g_m = 180 \mu\text{S}$ and $I_D = 10 \mu\text{A}$, the length is set to $L = 320 \text{ nm}$ to balance transconductance and output resistance.

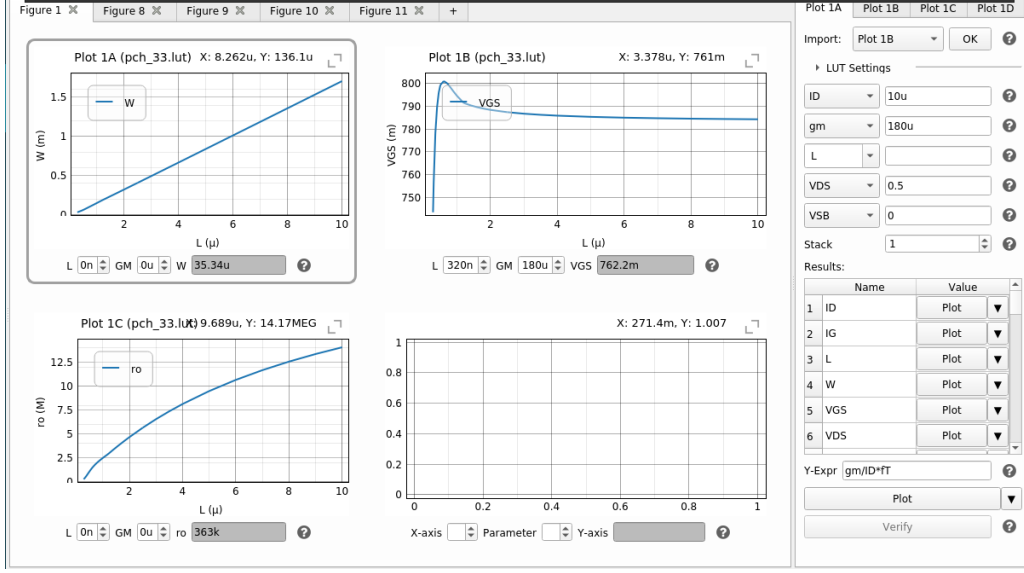


Figure 3: ADT Extraction for PMOS Input Pair

Extracted Parameters:

- $L = 320 \text{ nm}$
- $W = 35.34 \mu\text{m}$
- $V_{GS} = 762.2 \text{ mV}$
- $r_o = 363 \text{ k}\Omega$

2.3.2 PMOS Current Mirrors

For the PMOS top tail and bias current mirrors, V_{DS} is set to 0.625 V ($V_{DD}/4$) to provide sufficient output swing. To minimize noise and offset contributions, they are biased in strong inversion with $g_m/I_D = 10 \text{ V}^{-1}$ and a relatively long channel $L = 1 \mu\text{m}$. The branch current is set to $10 \mu\text{A}$.

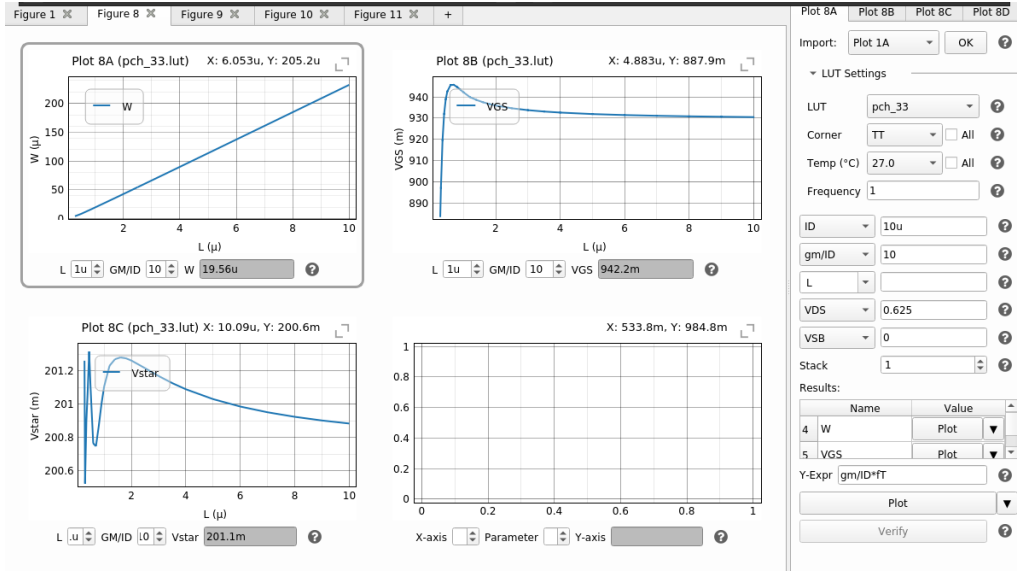


Figure 4: ADT Extraction for PMOS Current Mirrors

Extracted Parameters:

- $L = 1 \mu\text{m}$
- $W = 19.56 \mu\text{m}$
- $V_{GS} = 942.2 \text{ mV}$
- $V^* = 201.1 \text{ mV}$

2.3.3 NMOS Current Mirrors

The NMOS bottom current sinks are designed with identical constraints as the PMOS mirrors ($V_{DS} = 0.625 \text{ V}$, $g_m/I_D = 10 \text{ V}^{-1}$, $L = 1 \mu\text{m}$, $I_D = 10 \mu\text{A}$) to limit their noise injection and offset at the folding node.

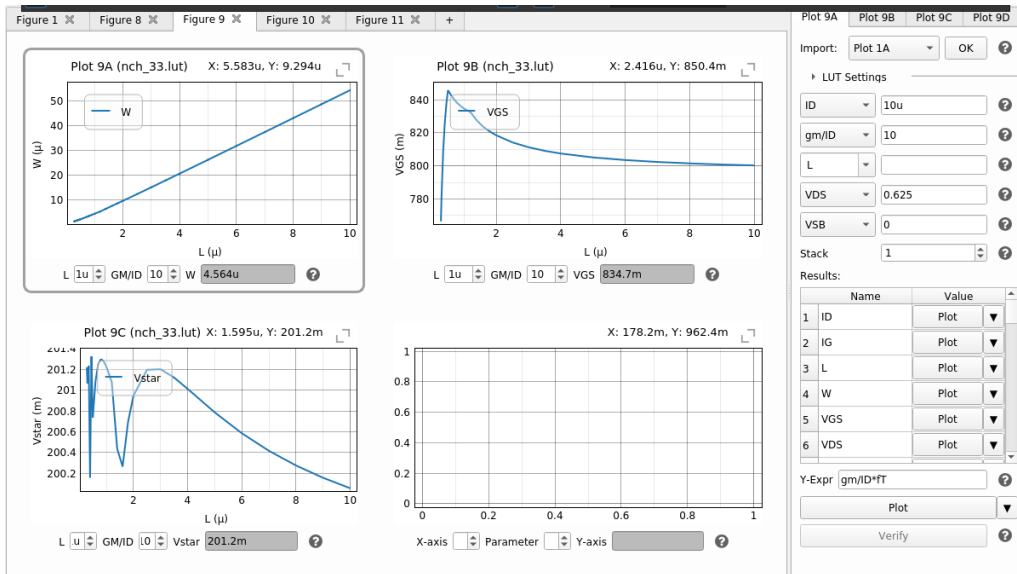


Figure 5: ADT Extraction for NMOS Current Mirrors

Extracted Parameters:

- $L = 1 \mu\text{m}$
- $W = 4.564 \mu\text{m}$
- $V_{GS} = 834.7 \text{ mV}$
- $V^* = 201.2 \text{ mV}$

2.3.4 Cascode PMOS

The PMOS cascode transistors do not contribute significantly to noise, allowing for moderate inversion biasing. The parameters are set to $g_m/I_D = 15 \text{ V}^{-1}$, $I_D = 10 \mu\text{A}$, and $V_{DS} = 0.625 \text{ V}$. A channel length of $L = 350 \text{ nm}$ is chosen over the standard 500 nm to secure sufficient output resistance (r_o) while maintaining a lower intrinsic gain to optimize the frequency response.

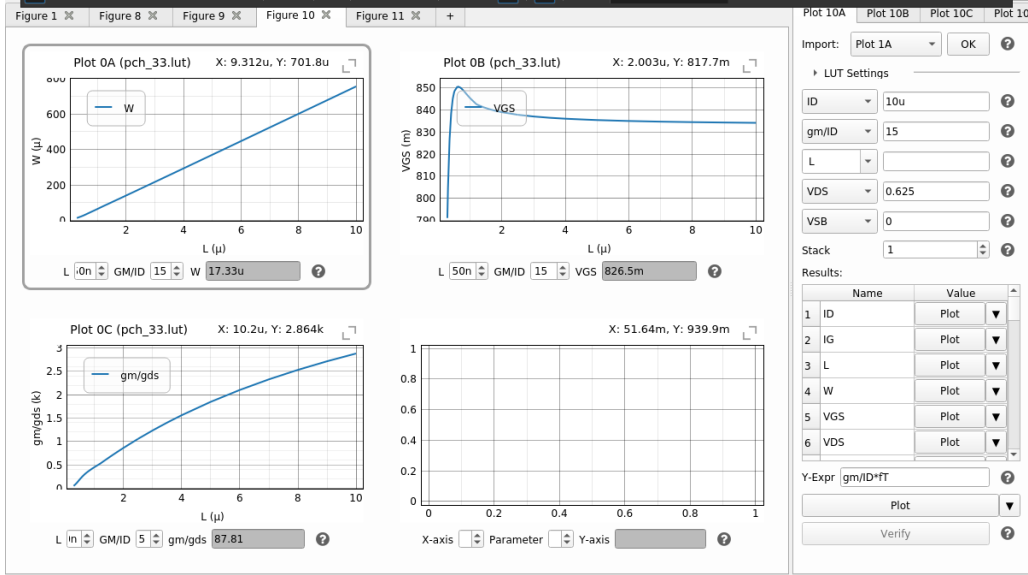


Figure 6: ADT Extraction for Cascode PMOS

Extracted Parameters:

- $L = 350 \text{ nm}$
- $W = 17.33 \mu\text{m}$
- $V_{GS} = 826.5 \text{ mV}$
- $g_m/g_{ds} = 87.81$

2.3.5 Cascode NMOS

The NMOS cascode transistors are sized using the same logic and constraints as their PMOS counterparts ($g_m/I_D = 15 \text{ V}^{-1}$, $L = 350 \text{ nm}$, $V_{DS} = 0.625 \text{ V}$).

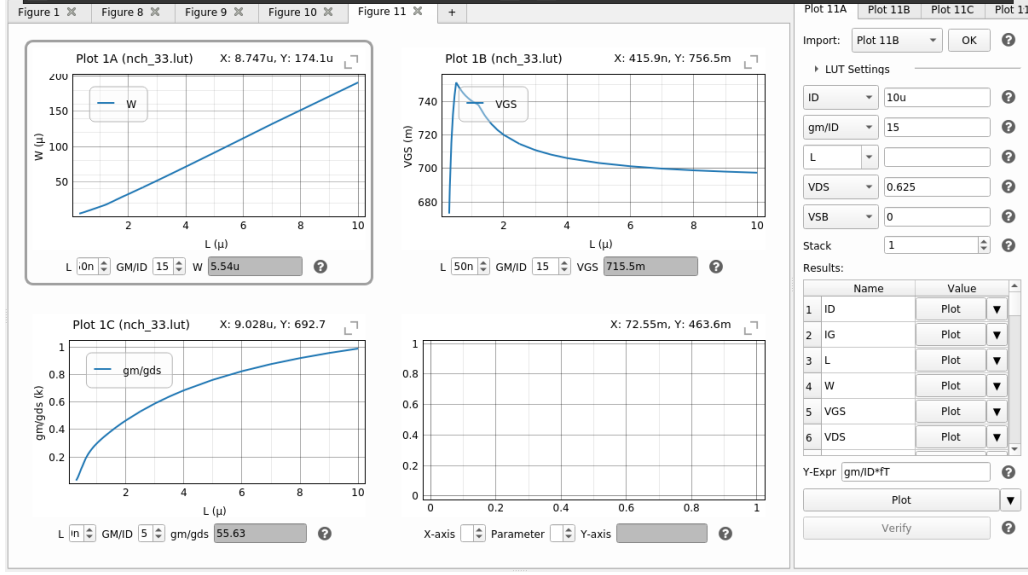


Figure 7: ADT Extraction for Cascode NMOS

Extracted Parameters:

- $L = 350 \text{ nm}$
- $W = 5.54 \mu\text{m}$
- $V_{GS} = 715.5 \text{ mV}$
- $g_m/g_{ds} = 55.63$

2.4 Cascode Biasing Voltages (V_{CASCN} and $V_{CASC P}$)

The fixed biasing voltages for the cascode transistors were calculated to ensure all devices, including the tail current sources, remain in the saturation region.

For the NMOS cascode bias (V_{CASCN}), the voltage must account for the V_{GS} of the NMOS cascode itself and the $V_{DS,sat}$ (V^*) of the bottom NMOS current sink:

$$V_{CASCN} \approx V_{GS,cascN} + V_{botN}^* \quad (6)$$

$$V_{CASCN} = 715.5 \text{ mV} + 201.2 \text{ mV} = 916.7 \text{ mV} \approx 0.92 \text{ V} \quad (7)$$

For the PMOS cascode bias ($V_{CASC P}$), the voltage drops from V_{DD} and must account for the $|V_{GS}|$ of the PMOS cascode and the V^* of the top PMOS tail current source:

$$V_{CASC P} \approx V_{DD} - |V_{GS,cascP}| - V_{topP}^* \quad (8)$$

$$V_{CASC P} = 2.5 \text{ V} - 0.8265 \text{ V} - 0.2011 \text{ V} = 1.4724 \text{ V} \approx 1.47 \text{ V} \quad (9)$$

3 Part 3: Open-Loop Simulation and Analysis

3.1 Schematic Implementations

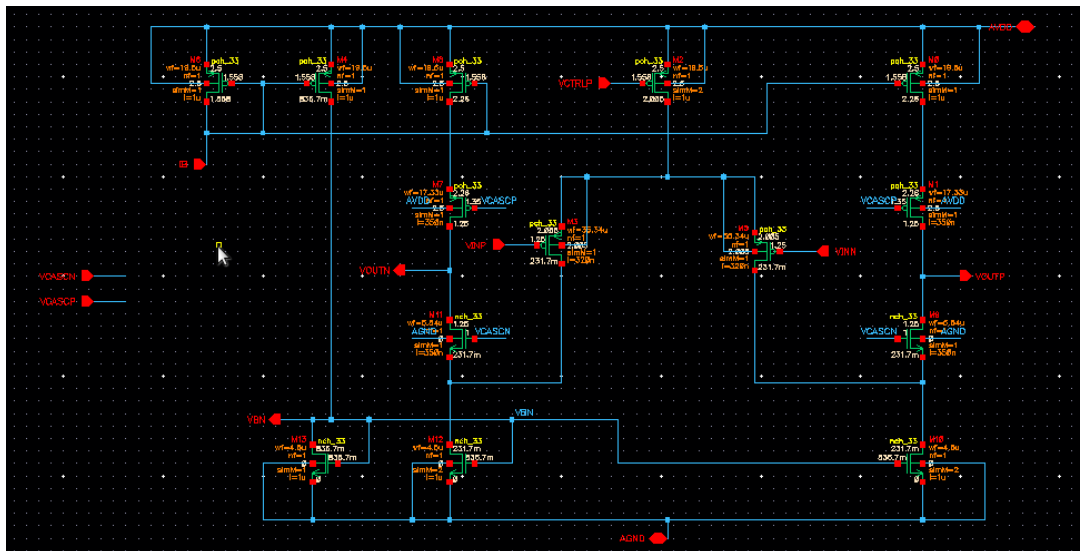


Figure 8: Fully-Differential Folded Cascode OTA Schematic

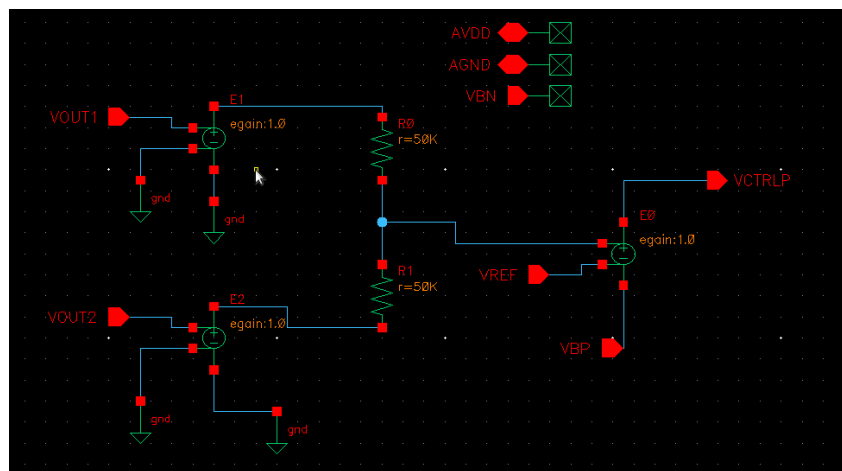


Figure 9: Behavioral CMFB Circuit Schematic

3.2 DC Operating Point Analysis

By iteratively tuning the bias voltages to ensure all transistors are in the saturation region and centered for maximum output swing, the fixed cascode biases were adjusted slightly from the hand calculations to $V_{CASCN} = 1.06$ V and $V_{CASCN} = 1.35$ V.

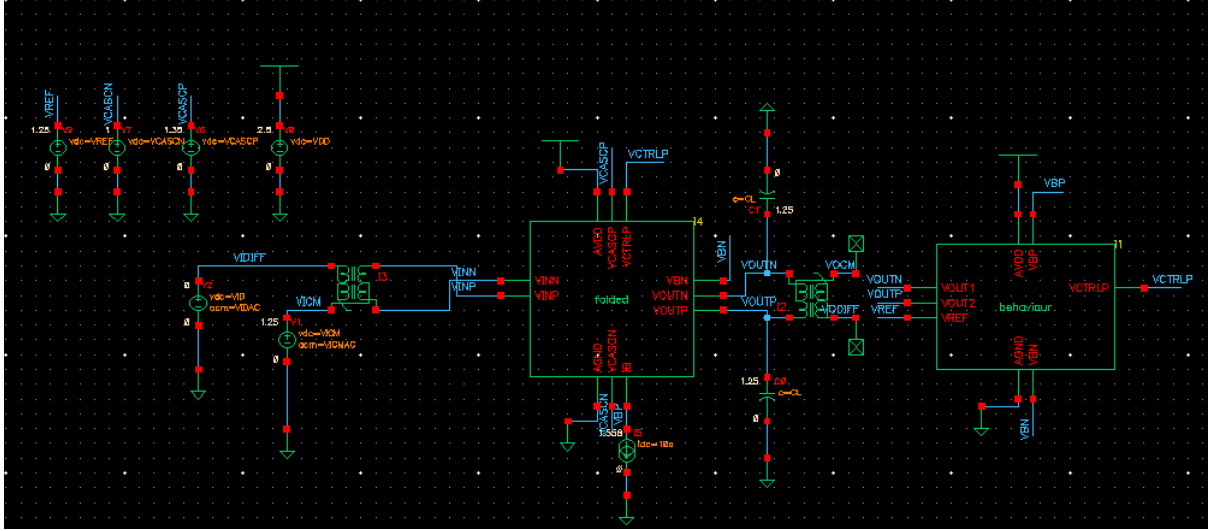


Figure 10: Open-Loop DC Testbench displaying 1.25 V output

3.2.1 DC Analysis Questions

What is the CM level at the OTA output?

As annotated on the testbench schematic, the common-mode output level settled at 1.25 V. This confirms that the behavioral CMFB loop is functioning correctly and has forced the output to track V_{REF} .

What are the differential input and output voltages of the error amplifier? What is the relation between them?

The differential input to the error amplifier is the difference between the sensed common-mode voltage and the reference voltage ($V_{CM} - V_{REF}$). Because the loop has reached steady-state equilibrium, this differential input is 0 V. The output of the error amplifier (V_{CTRLP}) is defined by the equation $V_{CTRLP} = A(V_{CM} - V_{REF}) + V_{BP}$. Since the differential input is 0 V, the output voltage is forced to equal V_{BP} .

3.3 Small-Signal AC Analysis

3.3.1 Initial Results and Design Modification

The initial AC simulation (Figure 11) yielded a DC gain of 64.98 dB, which met the 60 dB requirement but left minimal margin. To improve the open-loop gain by increasing the output resistance (r_o), the channel length (L) of all cascode transistors (M1, M7, M9, M11) was modified from 350 nm to 400 nm.

Test	Output	Nominal	Spec	Weight	Pass/Fail
FOLDEDFoLdEd:1	Ao	1.774k			
FOLDEDFoLdEd:1	Ao_db	64.98			
FOLDEDFoLdEd:1	BW	30.22k			
FOLDEDFoLdEd:1	UGF	54.12M			
FOLDEDFoLdEd:1	GBW	53.59M			
FOLDEDFoLdEd:1	PM	84.29			

Figure 11: Initial AC Results ($L_{casc} = 350$ nm) showing $A_v = 64.98$ dB

The subsequent simulation (Figures 12 and 13) demonstrated an improvement, achieving a DC gain of 69.84 dB and a Phase Margin of 84.05°, exceeding all target specifications.

Test	Output	Nominal	Spec	Weight	Pass/Fail
FOLDED:FoLdEd:1	Ao	3.104k			
FOLDED:FoLdEd:1	Ao_db	69.84			
FOLDED:FoLdEd:1	BW	17.6k			
FOLDED:FoLdEd:1	UGF	55.14M			
FOLDED:FoLdEd:1	GBW	54.62M			
FOLDED:FoLdEd:1	PM	84.05			

Figure 12: Modified AC Results ($L_{casc} = 400$ nm) showing $A_v = 69.84$ dB

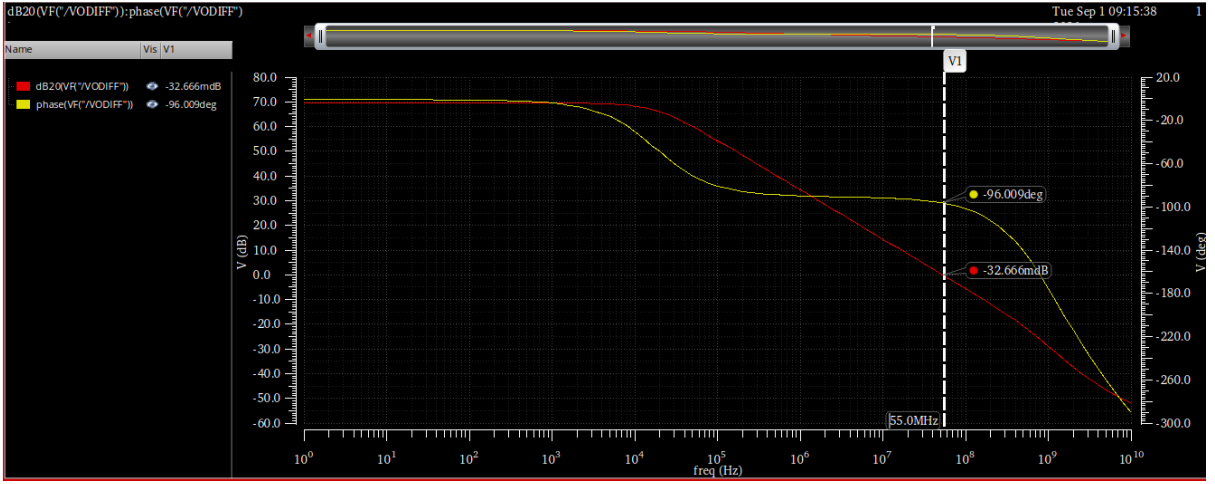


Figure 13: Open-Loop AC Bode Plot (Magnitude and Phase)

3.3.2 Hand Calculations vs. Simulation Comparison

The output resistance looking up into the PMOS cascode stack (R_{up}) is:

$$R_{up} \approx (g_{m,cascP} + g_{mb,cascP})r_{o,cascP}r_{o,topP} \quad (10)$$

$$R_{up} = (140.8\mu + 51.21\mu) \times \left(\frac{1}{0.992\mu} \right) \times \left(\frac{1}{2.697\mu} \right) \approx 71.7 \text{ M}\Omega \quad (11)$$

The output resistance looking down into the NMOS cascode stack and folding node (R_{down}) is:

$$R_{down} \approx (g_{m,cascN} + g_{mb,cascN})r_{o,cascN}(r_{o,botN} \parallel r_{o,inP}) \quad (12)$$

$$R_{down} = (140.9\mu + 38.44\mu) \times \left(\frac{1}{1.558\mu} \right) \times \left(\frac{1}{3.226\mu} \parallel \frac{1}{1.954\mu} \right) \approx 22.2 \text{ M}\Omega \quad (13)$$

The total output resistance (R_{out}) and DC gain (A_v) are calculated as:

$$R_{out} = R_{up} \parallel R_{down} = 71.7 \text{ M}\Omega \parallel 22.2 \text{ M}\Omega = 16.95 \text{ M}\Omega \quad (14)$$

$$A_v = g_{m1} \times R_{out} = 182.2 \mu\text{S} \times 16.95 \text{ M}\Omega = 3088 \text{ V/V} \quad (15)$$

$$A_{v,dB} = 20 \log_{10}(3088) \approx 69.79 \text{ dB} \quad (16)$$

Using the derived output resistance and the total load capacitance, the 3-dB Bandwidth (BW) is evaluated:

$$BW \approx \frac{1}{2\pi R_{out} C_L} = \frac{1}{2\pi(16.95 \text{ M}\Omega)(500 \text{ fF})} = 18.77 \text{ kHz} \quad (17)$$

The open-loop Gain-Bandwidth Product (GBW) is dictated by the input transconductance and load capacitance:

$$GBW \approx \frac{g_{m1}}{2\pi C_L} = \frac{182.2 \mu\text{S}}{2\pi(500 \text{ fF})} = 58.00 \text{ MHz} \quad (18)$$

Finally, the Phase Margin (PM) is estimated based on the dominant pole approximation and the non-dominant pole (f_{nd}) located at the internal folding node:

$$PM \approx 90^\circ - \tan^{-1} \left(\frac{GBW}{f_{nd}} \right) \quad \text{where} \quad f_{nd} \approx \frac{g_{m,cascN}}{2\pi C_{fold}} \quad (19)$$

As summarized in Table 2, the hand calculations match the simulated results.

Table 2: Small-Signal Parameters: Hand Calculations vs. Simulation

Parameter	Hand Calculation	Simulation
DC Gain (A_v) Linear	3088 V/V	3104 V/V
DC Gain (A_v) Decibels	69.79 dB	69.84 dB
Bandwidth (BW)	18.77 kHz	17.60 kHz
Gain-Bandwidth (GBW)	58.00 MHz	54.62 MHz
Phase Margin (PM)	$\approx 85^\circ - 88^\circ$	84.05°

4 Part 4: Transistor-Level CMFB Design

4.0.1 Current Budget

To maintain low power consumption, the current budget for the entire CMFB circuit was constrained to $23 \mu\text{A}$ (less than 50% of the main $50 \mu\text{A}$ OTA current). The current was distributed as follows:

- $6.5 \mu\text{A}$ allocated to each of the three PMOS buffer branches (two for the output nodes, one dummy for the reference).
- $10 \mu\text{A}$ allocated to the NMOS error amplifier tail current, splitting into $5 \mu\text{A}$ per branch.

A sensing resistor value of $R_{CM} = 150 \text{ k}\Omega$ was selected. This ensures that the peak current drawn during a maximum voltage swing remains below the $6.5 \mu\text{A}$ buffer bias, preventing the PMOS source followers from turning off.

4.0.2 Maximizing Output Swing (Centering V_{REF})

The maximum output voltage is limited by the top PMOS current sources ($V_{DD} - V_{OV} - V_{SG} \approx 1.45 \text{ V}$). The minimum output voltage is same ($2V_{ov}$) ($\approx 0.4 \text{ V}$). To achieve a symmetrical maximum voltage swing, the common-mode reference (V_{REF}) was calculated as the midpoint between these two extremes:

$$V_{REF(opt)} = \frac{V_{OUT,max} + V_{OUT,min}}{2} = \frac{1.45 \text{ V} + 0.4 \text{ V}}{2} = 0.925 \text{ V} = 925 \text{ mV} \quad (20)$$

To accommodate this, a third "dummy" PMOS buffer branch was integrated into the CMFB schematic. This dummy buffer shifts the 925 mV reference voltage up by the same V_{GS} drop as the sensing buffers, ensuring a balanced comparison at the error amplifier's inputs.

4.1 Analog Designer's Toolbox (ADT) Sizing

The devices in the CMFB block were explicitly sized using ADT to meet the current budget.

4.1.1 PMOS Input Buffers & Dummy Buffer

To ensure voltage tracking, the PMOS source followers were sized with $g_m/I_D = 15 \text{ V}^{-1}$, $L = 1 \mu\text{m}$, and $V_{DS} = 1.25 \text{ V}$ for a branch current of $I_D = 6.5 \mu\text{A}$.

Name		TT-27.0
4	W	41.4u
5	VDB	1.25
6	VGB	846.2m
7	VGS	846.2m
8	VDS	1.25
9	VSB	0

Figure 14: ADT Extraction for PMOS Input Buffers

Extracted Parameters: $W = 41.4 \mu\text{m}$, $V_{GS} = 846.2 \text{ mV}$.

4.1.2 PMOS Current Sources (VBP)

The PMOS current sources biasing the buffers were sized identically to the main OTA mirrors, utilizing $g_m/I_D = 10 \text{ V}^{-1}$, $L = 1 \mu\text{m}$, and $V_{DS} = 1.25 \text{ V}$ at $I_D = 6.5 \mu\text{A}$.

ID	6.5u	?
gm/ID	10	?
L	1u	?
VDS	1.25	?
VSB	0	?
Stack	1	?

Results:

	Name	TT-27.0
4	W	12.52u
5	VDB	1.25
6	VGB	943.5m
7	VGS	943.5m
8	VDS	1.25
9	VSB	0

Figure 15: ADT Extraction for PMOS Current Sources

Extracted Parameters: $W = 12.52 \mu\text{m}$, $V_{GS} = 943.5 \text{ mV}$.

4.1.3 PMOS Diode-Connected Loads

The active loads for the error amplifier were sized with $g_m/I_D = 10 \text{ V}^{-1}$, $L = 1 \mu\text{m}$, and $V_{DS} = 0.83 \text{ V}$ (approx. $V_{DD}/3$) at a split current of $I_D = 5 \mu\text{A}$.

► LUT Settings

ID	5u	?
gm/ID	10	?
L	1u	?
VDS	0.83	?
VSB	0	?
Stack	1	?

Results:

	Name	TT-27.0
1	ID	5u
2	IG	N/A
3	L	1u
4	W	9.72u
5	VDB	830m
6	VGB	943.5m

Figure 16: ADT Extraction for PMOS Diode Loads

Extracted Parameters: $W = 9.72 \mu\text{m}$, $V_{GS} = 943.5 \text{ mV}$.

4.1.4 NMOS Differential Input Pair

The input pair of the error amplifier (V_{CM} , V_{REF}) was sized with $g_m/I_D = 15 \text{ V}^{-1}$, $L = 1 \mu\text{m}$, and $V_{DS} = 0.83 \text{ V}$ for $I_D = 5 \mu\text{A}$.

ID: 5u
 gm/ID: 15
 L: 1u
 VDS: 0.83
 VSB: 0
 Stack: 1

Results:

	Name	TT-27.0
1	ID	5u
2	IG	N/A
3	L	1u
4	W	7.24u
5	VDB	830m
6	VGB	741.8m

Figure 17: ADT Extraction for NMOS Diff Pair

Extracted Parameters: $W = 7.24 \mu\text{m}$, $V_{GS} = 741.8 \text{ mV}$.

4.1.5 NMOS Tail Current Source (VBN)

The bottom error amplifier tail was matched to the OTA NMOS sinks using $g_m/I_D = 10 \text{ V}^{-1}$, $L = 1 \mu\text{m}$, and $V_{DS} = 0.83 \text{ V}$ to pull a total of $I_D = 10 \mu\text{A}$.

ID: 10u
 gm/ID: 10
 L: 1u
 VDS: 0.83
 VSB: 0
 Stack: 1

Results:

	Name	TT-27.0
1	ID	10u
2	IG	N/A
3	L	1u
4	W	4.56u
5	VDB	830m
6	VGB	835.5m

Y-Expr: gm/ID*ft

Figure 18: ADT Extraction for NMOS Tail

Extracted Parameters: $W = 4.56 \mu\text{m}$, $V_{GS} = 835.5 \text{ mV}$.

4.2 Schematic Implementation

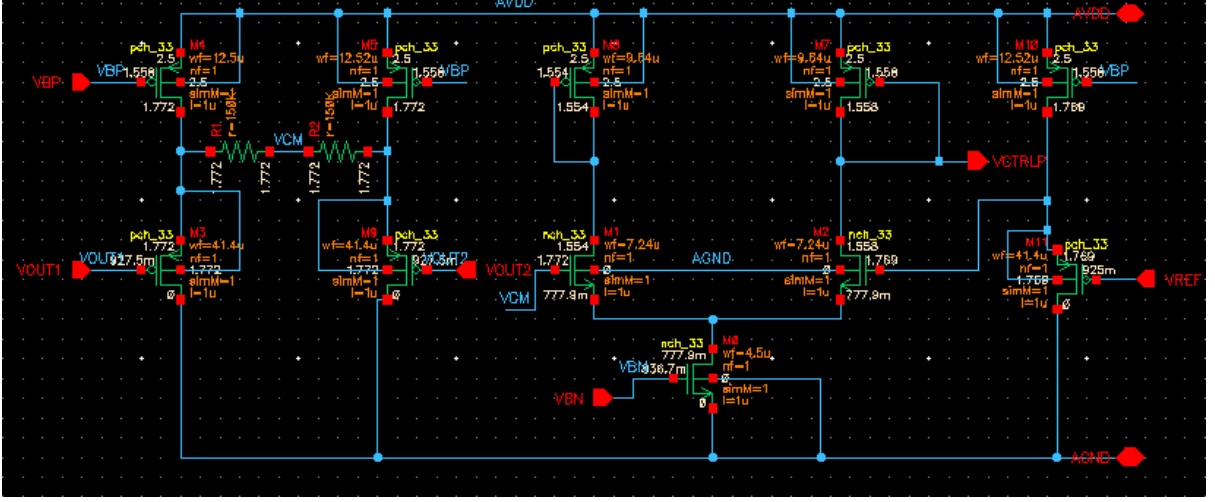


Figure 19: Transistor-Level CMFB Schematic (with Dummy Buffer Branch)

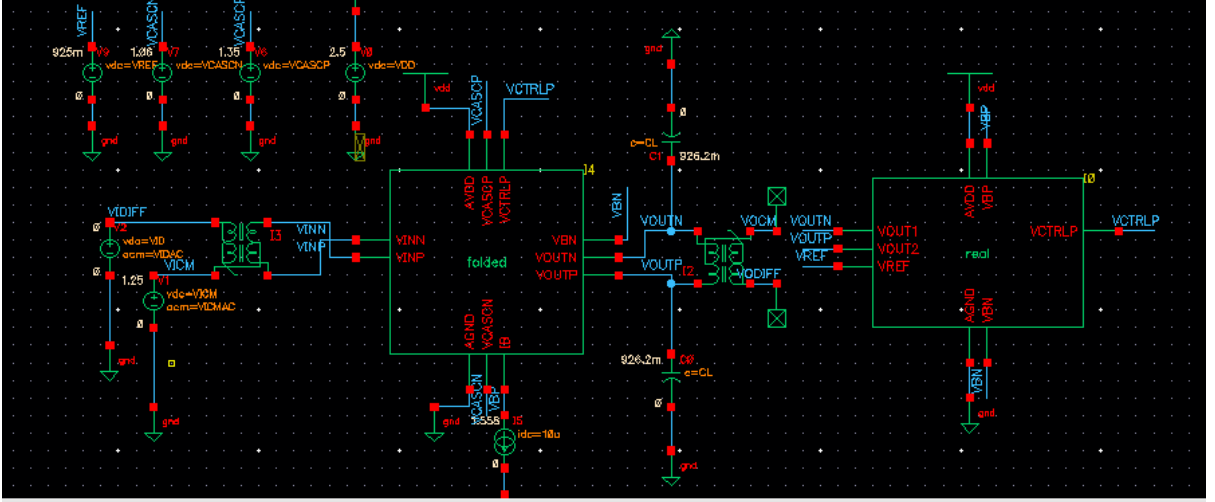


Figure 20: Open-Loop Testbench emphasizing the 925 mV V_{REF}

4.3 DC Operating Point Analysis

4.3.1 DC Analysis Summary

With the transistor-level feedback loop operational and the V_{REF} source set to 925 mV, the common-mode output level (V_{OUT1} , V_{OUT2}) settled at 926.2 mV (annotated in Figure 20).

By centering the output at 926.2 mV rather than 1.25 V, the NMOS cascode transistors experience a slightly reduced drain-source voltage (V_{DS}).

4.3.2 What are the differential input and output voltages of the error amplifier? What is the relation between them?

The differential input is the difference between the sensed common-mode voltage and the reference voltage ($V_{CM} - V_{REF}$). Because the feedback loop is stable and settled, this difference is approximately 0.

The output is the control voltage (V_{CTRLP}) that biases the top PMOS current sources of the main OTA.

The relationship is based on negative feedback. The error amplifier multiplies the input difference by its gain to generate V_{CTRLP} . If the output common-mode voltage shifts, this creates a difference at the input. The amplifier reacts by adjusting V_{CTRLP} to change the OTA's current, forcing the output back to V_{REF} (925 mV).

4.3.3 Transistor Operating Parameters

The simulated DC operating points for both the main Folded Cascode OTA and the Transistor-Level CMFB circuit are summarized in Tables 3 and 4.

Table 3: Simulated OP Parameters: Main Folded Cascode OTA

Device	Type	Function	I_D (μ A)	$ V_{GS} $ (mV)	$ V_{DS} $ (mV)	$ V_{DS,sat} $ (mV)	$ V_{TH} $ (mV)	g_m (μ S)	g_{ds} (μ S)	Region
M0	PMOS	Top CMFB Source	9.64	941.8	218.8	154.6	783.9	95.91	2.699	2
M1	PMOS	PMOS Cascode	9.64	931.2	1026.0	116.7	863.9	140.8	0.994	2
M2	PMOS	Tail Source	20.03	943.3	494.4	155.6	783.9	200.8	0.799	2
M3	PMOS	Input Pair	10.01	755.6	1753.0	81.64	751.1	182.2	1.954	2
M4	PMOS	Bias Diode	10.13	941.8	1663.0	154.6	783.9	102.2	0.155	2
M5	PMOS	Input Pair	10.01	755.6	1753.0	81.64	751.1	182.2	1.954	2
M6	PMOS	Bias Diode	10.00	941.8	941.8	154.6	783.9	100.9	0.229	2
M7	PMOS	PMOS Cascode	9.64	931.2	1026.0	116.7	863.9	140.8	0.994	2
M8	PMOS	Top CMFB Source	9.64	941.8	218.8	154.6	783.9	95.91	2.699	2
M9	NMOS	NMOS Cascode	9.64	807.2	1002.0	109.3	766.4	140.9	1.557	2
M10	NMOS	Bottom Sink	19.65	836.7	252.8	173.2	687.3	193.6	3.223	2
M11	NMOS	NMOS Cascode	9.64	807.2	1002.0	109.3	766.4	140.9	1.557	2
M12	NMOS	Bottom Sink	19.65	836.7	252.8	173.2	687.3	193.6	3.223	2
M13	NMOS	Bias Diode	10.13	836.7	836.7	173.2	687.3	100.2	0.330	2

Table 4: Simulated OP Parameters: Transistor-Level CMFB Block

Device	Type	Function	I_D (μ A)	$ V_{GS} $ (mV)	$ V_{DS} $ (mV)	$ V_{DS,sat} $ (mV)	$ V_{TH} $ (mV)	g_m (μ S)	g_{ds} (μ S)	Region
M1	NMOS	Diff Pair Input (V_{CM})	5.18	1064.0	520.0	117.3	1008.0	77.23	0.235	2
M2	NMOS	Diff Pair Input (V_{REF})	5.02	1062.0	523.0	116.0	1008.0	75.56	0.229	2
M0	NMOS	Tail Current Source	10.20	836.7	1034.0	173.2	687.3	100.7	0.304	2
M3	PMOS	Output Buffer (Bot)	6.31	843.3	2098.0	93.2	783.9	96.3	0.101	2
M9	PMOS	Output Buffer (Bot)	6.31	843.3	2098.0	93.2	783.9	96.4	0.101	2
M4	PMOS	Output Buffer (Top)	6.30	941.8	402.0	154.6	783.9	63.6	0.324	2
M5	PMOS	Output Buffer (Top)	6.31	941.8	401.9	154.6	783.9	63.7	0.324	2
M8	PMOS	Active Load (Diode)	5.18	946.3	946.3	157.7	783.9	51.3	0.118	2
M7	PMOS	Active Load (Mirror)	5.02	943.3	943.3	155.6	783.9	50.3	0.115	2

4.4 Small-Signal AC Analysis

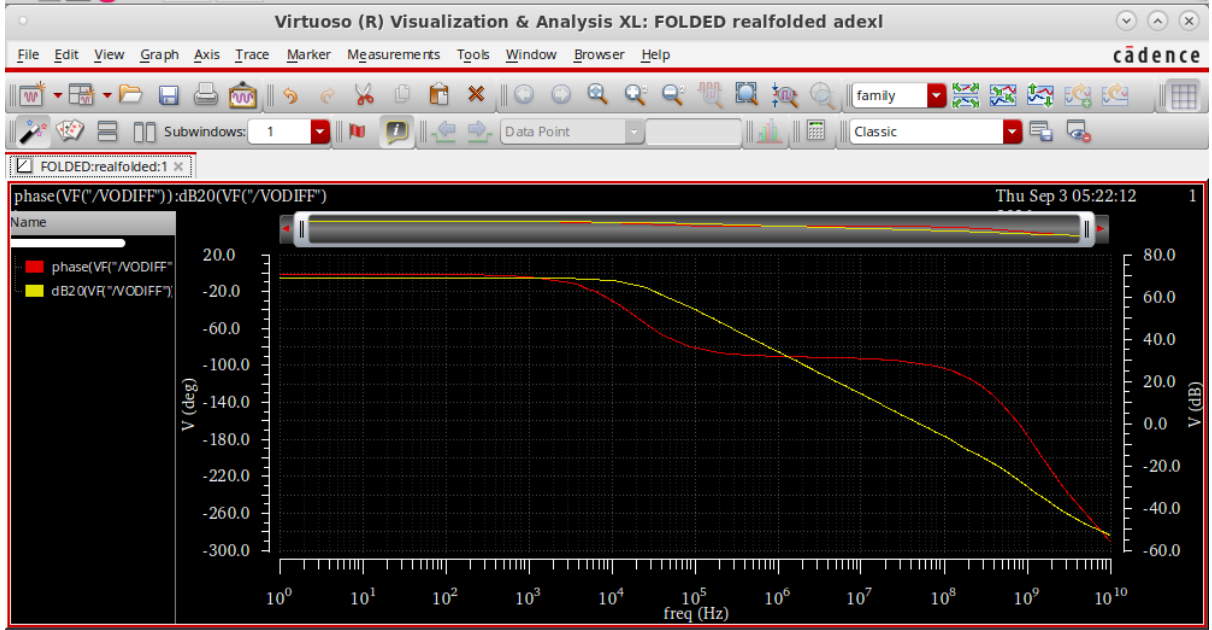


Figure 21: Open-Loop AC Bode Plot with Optimized Transistor-Level CMFB

Test	Output	Nominal	Spec	Weight	Pass/Fail
FOLDEDrealfolded:1	Ao	2.906k			
FOLDEDrealfolded:1	Ao_db	69.27			
FOLDEDrealfolded:1	BW	18.24k			
FOLDEDrealfolded:1	UGF	53.05M			
FOLDEDrealfolded:1	GBW	53.01M			
FOLDEDrealfolded:1	PM	83.42			
FOLDEDrealfolded:1	phase(VF(*/VODIFF))				
FOLDEDrealfolded:1	dB20(VF(*/VODIFF))				

Figure 22: Extracted AC Simulation Results ($V_{REF} = 925$ mV)

The overall performance remains robust and exceeds all target specifications. The slight adjustment of the DC operating point to 926.2 mV naturally resulted in a minor decrease in DC gain (from 69.84 dB to 69.27 dB) due to the slight reduction in NMOS output resistance (r_o) at lower V_{DS} values. However, the 69.27 dB gain still exceeds the 60 dB specification, and the Phase Margin remains stable at 83.42°.

4.5 Closed-Loop Testbench and DC Operating Point

The OTA was connected in a fully-differential capacitive feedback configuration. The closed-loop testbench architecture, with annotated DC operating voltages, is presented in Figure 23.

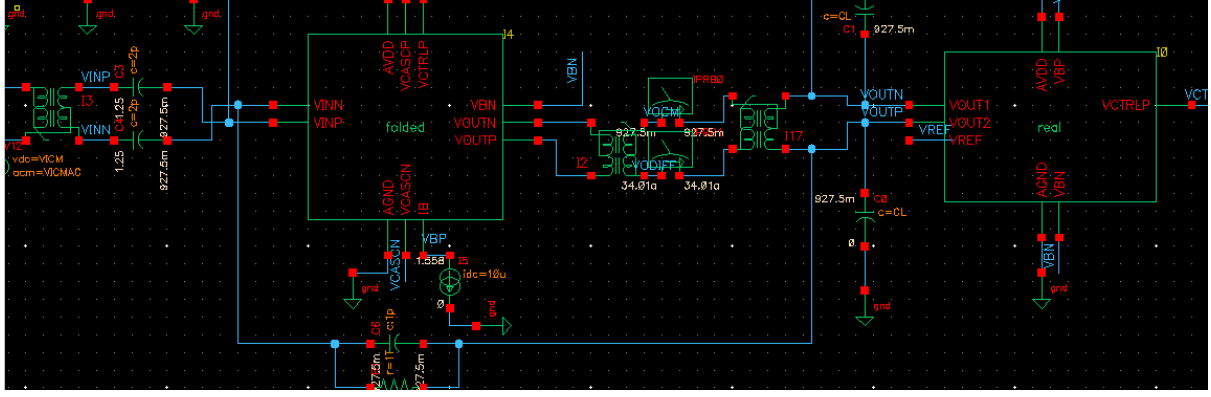


Figure 23: Closed-Loop Testbench emphasizing DC node voltages (External Input = 1.25 V, Internal Nodes ≈ 927.5 mV)

As shown in Figure 23, the external DC common-mode voltage is driven by the ideal input sources at 1.25 V. Because the 2 pF input capacitors act as open circuits to DC signals, the internal OTA input nodes are isolated from this external DC source.

To establish the correct DC operating point for the OTA's input pair, a large-value DC feedback resistor network is utilized to provide a DC path between the output and input terminals. As annotated on the schematic, this configuration forces the internal input nodes to track the output common-mode voltage. Consequently, both the input and output terminals of the OTA settle at approximately V_{REF} (927.5 mV). The Transistor-Level CMFB loop effectively dictates the DC operating point for the entire internal closed-loop structure.

4.6 Differential AC Closed-Loop Response

To verify the closed-loop bandwidth and gain, an AC simulation was executed (Figures 24 and 25). The capacitive feedback network achieved the target flat mid-band gain of 1.998 V/V (approximately 6 dB). The -3 dB closed-loop bandwidth (BW) was measured at 8.822 MHz, yielding a closed-loop Gain-Bandwidth Product ($GBW \approx A_{cl} \times BW$) of 17.62 MHz.

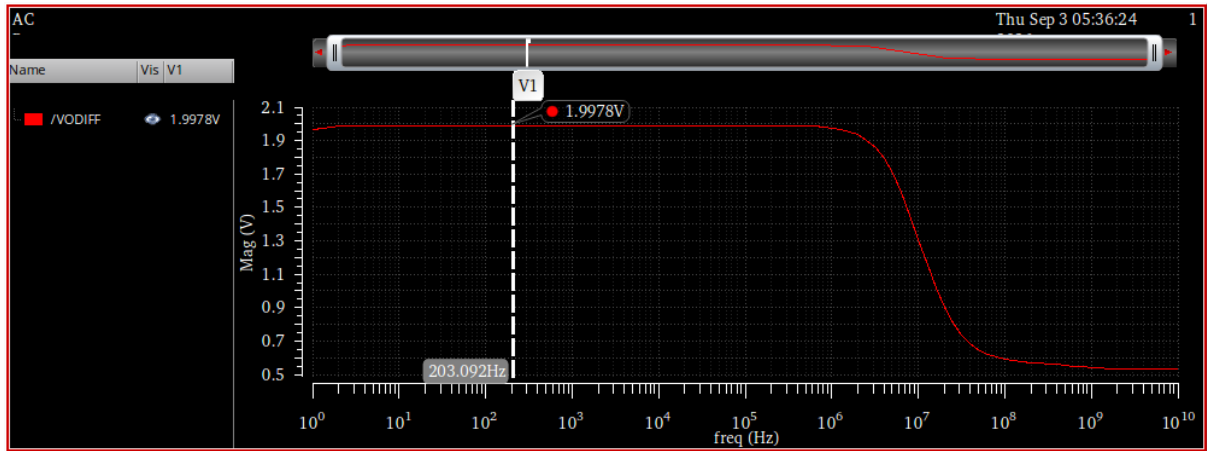


Figure 24: Closed-Loop Differential AC Bode Plot showing flat mid-band gain (1.9978 V/V)

FOLDED:close dloop:1	/VODIFF				
FOLDED:close dloop:1	BW	8.822M			
FOLDED:close dloop:1	Ao	1.998			
FOLDED:close dloop:1	(Ao * BW)	17.62M			

Figure 25: Extracted AC Closed-Loop Parameters ($A_v = 1.998\text{V/V}$, $GBW = 17.62\text{MHz}$)

4.7 Stability (STB) Analysis

To evaluate the stability margins of both the differential and common-mode feedback loops, an STB analysis was performed using a ‘cmdmprobe’ inserted directly at the output terminals (as previously visible in Figure 23).

4.7.1 Common-Mode and Differential STB Results

The simulated STB loop gain and phase plots for the common-mode loop are presented in Figures 26 and 27. The corresponding plots and extracted parameters for the differential loop are presented in Figures 28 and 29.

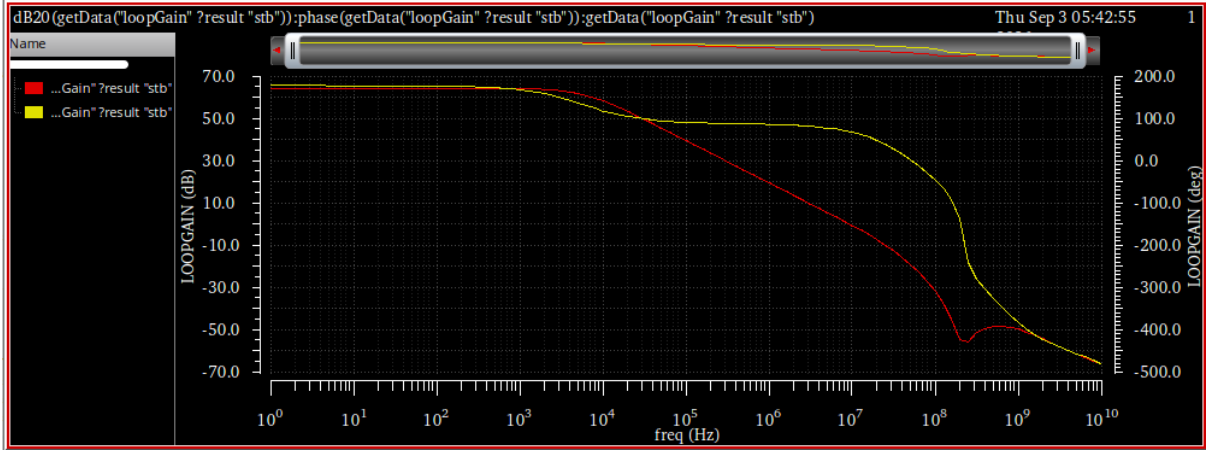


Figure 26: Common-Mode STB Loop Gain and Phase Plot

FOLDED:close dloop:1	unityGainFreq(getData("loopGai...	9.689M			
FOLDED:close dloop:1	gain	1.715k			
FOLDED:close dloop:1	GBW	9.885M			
FOLDED:close dloop:1	bw	5.764k			
FOLDED:close dloop:1	(phaseMargin(getData("loopGai...	70.72			

Figure 27: Extracted Common-Mode STB Parameters ($PM = 70.72^\circ$)

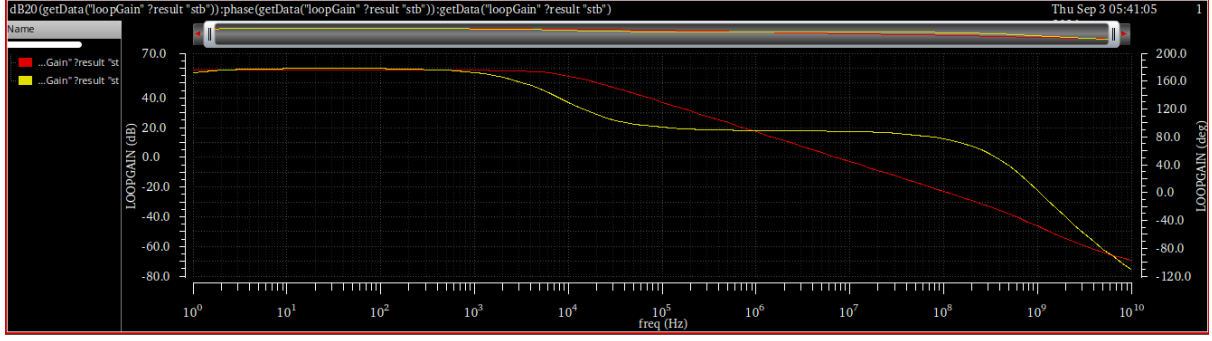


Figure 28: Differential STB Loop Gain and Phase Plot

FOLDED:closedloop:1	unityGainFreq(getData("loopGai...	7.713M			
FOLDED:closedloop:1	gain	915.7			
FOLDED:closedloop:1	GBW	7.574M			
FOLDED:closedloop:1	bw	8.271k			
FOLDED:closedloop:1	(phaseMargin(getData("loopGai...	89.15			

Figure 29: Extracted Differential STB Parameters ($PM = 89.15^\circ$)

4.7.2 Stability Commentary and Open-Loop Comparison

Differential vs. Common-Mode Pole Dynamics:

Comparing the two loops shows differences in their pole dynamics, though both are stable. The differential loop has a single pole at the output because of C_L . Therefore, its phase margin is high, about 89.15° .

The common-mode loop naturally exhibits a lower phase margin because the CMFB network introduces a secondary internal pole at the gate of the top PMOS current sources (the V_{CTRLP} node). The combination of the output node pole and this internal CMFB pole creates a decrease in PM.

Closed-Loop vs. Open-Loop Correlation:

The Loop Gain (LG) should be:

$$LG = A_{ol} \times \beta \approx 2906 \times \frac{1}{3} \approx 968 \text{ V/V} \quad (21)$$

The STB simulation confirms this relationship, reporting a measured differential Loop Gain of **915.7 V/V** (the slight reduction is expected due to the parasitic input capacitance of the OTA lowering the actual effective β).

Furthermore, while the unloaded open-loop GBW was 53.01 MHz, the STB simulation reports a Loop GBW of 7.57 MHz. This reduction directly reflects the capacitive loading effect of the feedback network. The addition of the 2 pF and 1 pF capacitors significantly increases the effective load capacitance at the output nodes, shifting the dominant pole to a lower frequency .

5 Part 6: Closed-Loop Simulation (Transient Analysis)

5.1 1. Differential Input Pulse Response

To evaluate the time-domain stability and settling time of the fully-differential closed-loop amplifier, a transient analysis was performed using the optimized 925 mV CMFB architecture.

5.1.1 Test Setup

The generic closed-loop testbench was utilized. The common-mode input (V_{ICM}) was held at a constant 500 mV DC. The differential input source (V_{IDIFF}) was configured to generate a 100 mV step pulse with the following parameters: initial value = 0 V, pulse value = 100 mV, delay = 1 μ s, rise/fall times = 10 ns, pulse width = 1 μ s, and period = 2 μ s. The simulation was executed for 3 μ s with a maximum time step of 10 ns.

The transient waveforms for the input nodes (V_{INP} , V_{INN}), the output nodes (V_{OUTP} , V_{OUTN}), and the output common-mode node (V_{OCM}) are overlaid in Figure 30.

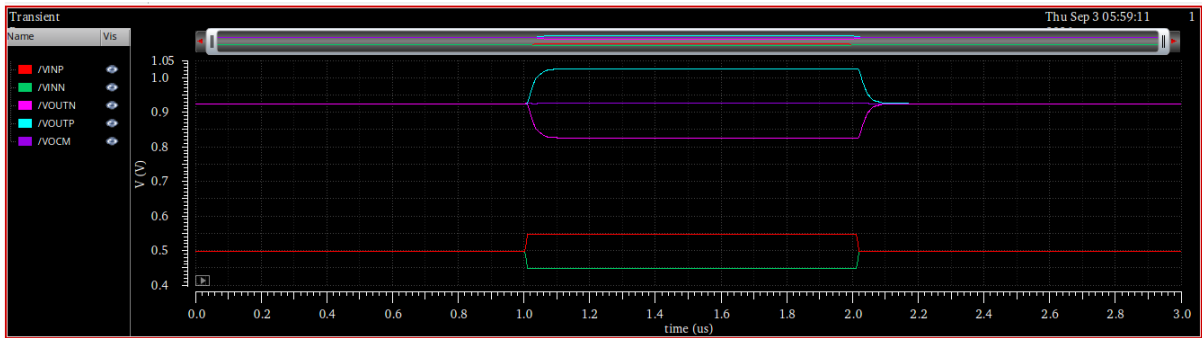


Figure 30: Transient Response to a 100 mV Differential Step ($V_{OCM} \approx 925$ mV)

5.1.2 Stability and Ringing Analysis

Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

As observed in Figure 30, there is no ringing on the differential output nodes (V_{OUTP} and V_{OUTN}). The voltages transition to their final settled values. This correlates with the final differential loop STB analysis, which demonstrated a stable phase margin of 89.15°.

Furthermore, VOCM stays nearly constant around 925 mV during the differential step. Therefore, the CMFB loop maintains the output common-mode level and no noticeable CM ringing is observed.

5.1.3 1% Settling Time Optimization

Calculate the 1% settling time and compare it to the required specification.

To satisfy the 100 ns specification, a final design optimization was performed: the main input pair was pushed slightly deeper into weak inversion by increasing its g_m/I_D target.

This change increased the input transconductance (g_m) and the overall Gain-Bandwidth Product, effectively decreasing the closed-loop time constant.

Following this optimization, the transient calculator measured a final 1% settling time of 98.09 ns (Figure 31), fulfilling the target specification.

Test	Output	Nominal	Spec	Weight	Pass/Fail
FOLDEDx:closedloop:1	/MNP				
FOLDEDx:closedloop:1	/MNN				
FOLDEDx:closedloop:1	/VOUTN				
FOLDEDx:closedloop:1	/VOUTP				
FOLDEDx:closedloop:1	/VOCM				
FOLDEDx:closedloop:1	(settlingTime(v("/VOUTP"?result...	98.09n			

Figure 31: ADE Transient Calculator measurement confirming 98.09 ns settling time

5.2 2. Common-Mode Input Pulse Response

To evaluate the stability and rejection capabilities of the Transistor-Level CMFB loop, a common-mode transient shock was applied to the closed-loop amplifier.

5.2.1 Test Setup

The differential input source (V_{IDIFF}) was disabled (set to 0 V DC). The common-mode input source (V_{ICM}) was changed to a pulse source to apply a 100 mV step. The pulse parameters were: delay = 1 μ s, rise/fall times = 10 ns, pulse width = 1 μ s, and period = 2 μ s. The simulation was run for 3 μ s.

The transient waveforms for V_{INP} , V_{INN} , V_{OUTP} , V_{OUTN} , and V_{OCM} are overlaid in Figure 32.

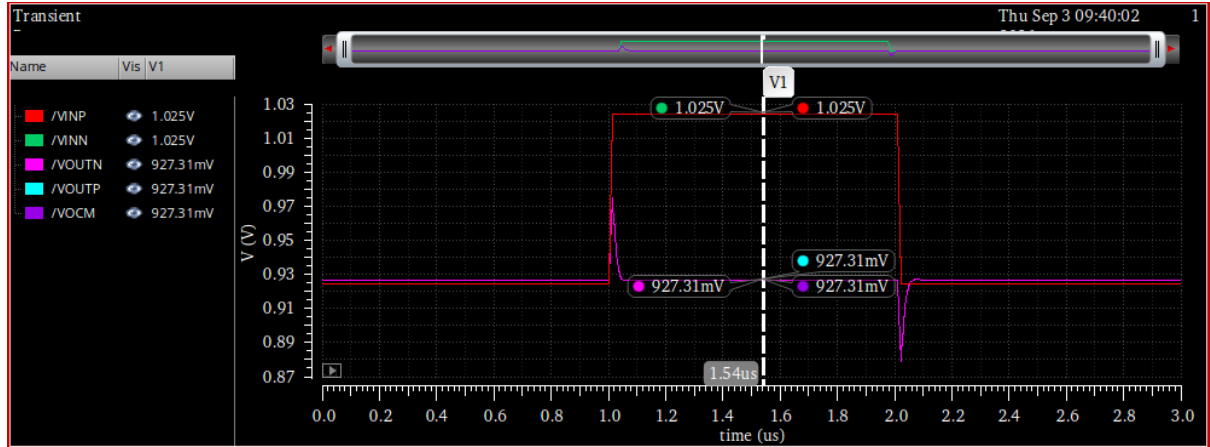


Figure 32: Transient Response to a 100 mV Common-Mode Input Step (V_{OCM} recovery to 925 mV)

5.2.2 Stability and Ringing Analysis

Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

As shown in Figure 32, a quick voltage spike appears on all output nodes at 1.0 μ s. This happens because the fast input signal leaks through the large feedback capacitors.

The CMFB loop instantly reacts and smoothly pulls the common-mode voltage (V_{OCM}) back to the 925 mV baseline. Because the signal settles without any ringing or oscillation, it proves the common-mode loop is highly stable, matching the 70.72° phase margin we measured earlier.

Finally, V_{OUTP} and V_{OUTN} overlap perfectly during the entire event. This shows that the differential loop completely ignores the common-mode jump, proving the amplifier has excellent Common-Mode Rejection Ratio (CMRR) and that the two loops do not interfere with each other.

5.3 3. Output Swing and Large-Signal Linearity

To verify the large-signal handling, maximum output swing, and DC isolation of the fully-differential amplifier, a large-signal sinusoidal transient analysis was conducted using the final optimized CMFB architecture.

5.3.1 Test Setup and DC Isolation

A test condition was applied to demonstrate the AC-coupled nature of the feedback network: the input common-mode voltage (V_{ICM}) was set to 0 V. The differential input source (V_{IDIFF}) was configured to generate a 100 kHz sine wave with a 600 mV peak-to-peak amplitude. The transient simulation was executed for $30\ \mu\text{s}$ (capturing three full periods).

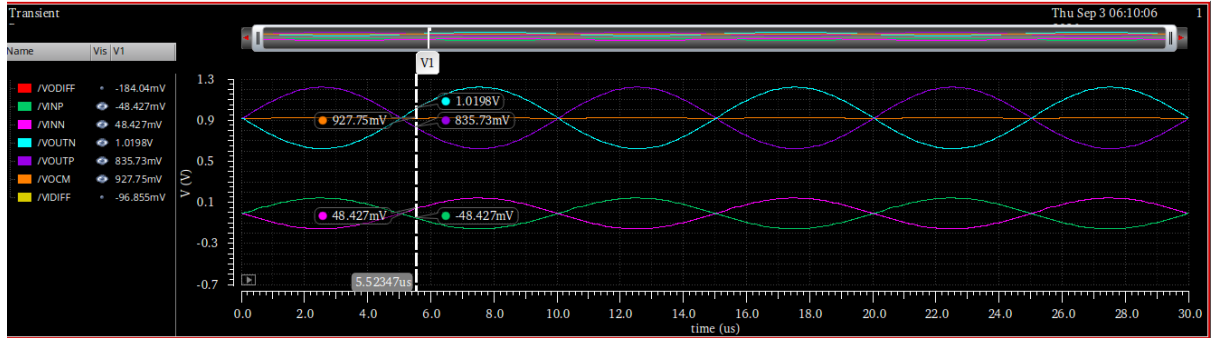


Figure 33: Single-Ended Transient Response with $V_{ICM} = 0\text{ V}$ and $V_{OCM} \approx 927.75\text{ mV}$

As shown in Figure 33, the input nodes (V_{INP} , V_{INN}) swing symmetrically around the 0 V input DC level. However, because the large input capacitors block the DC path, the output common-mode voltage (V_{OCM}) is isolated from this 0 V source. The CMFB loop holds the output center point at the optimized $\approx 927.75\text{ mV}$ baseline.

5.3.2 Waveform Linearity and Full-Scale Swing

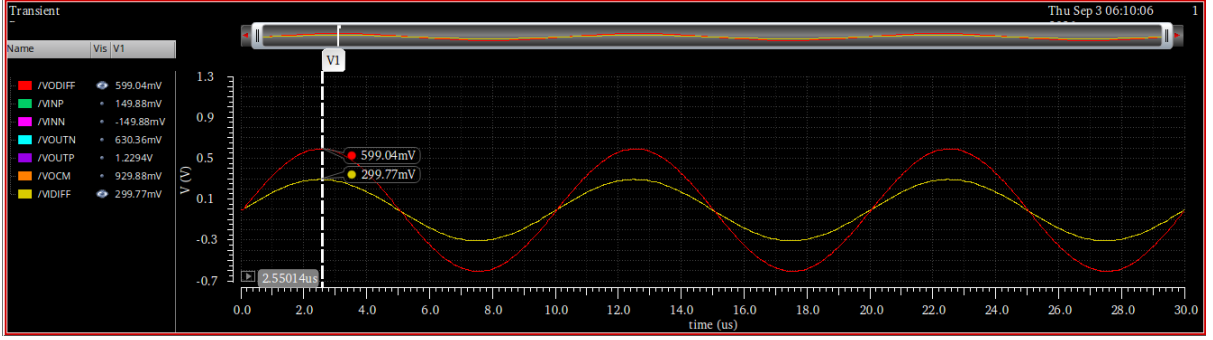


Figure 34: Differential Input vs. Differential Output Waveforms ($1.2 V_{p-p}$ output swing)

5.3.3 Closed-Loop Gain Calculation

To calculate the time-domain closed-loop gain under maximum swing conditions, the peak-to-peak voltages were extracted using the ADE calculator (Figure 35).

FOLDED:closedloop:1	peakToPeak(V("/VIDIFF"?result "...	599.9m			
FOLDED:closedloop:1	peakToPeak(V("/VIDIFF"?result "...	1.198			

Figure 35: ADE Calculator Extraction of Maximum Peak-to-Peak Differential Swings

The differential input swing was 599.9 mV_{p-p} , resulting in a full-scale differential output swing of $1.198 V_{p-p}$. The closed-loop time-domain gain (A_{cl}) is calculated as:

$$A_{cl} = \frac{V_{ODIFF(p-p)}}{V_{IDIFF(p-p)}} = \frac{1.198 \text{ V}}{0.5999 \text{ V}} \approx 1.997 \text{ V/V} \quad (22)$$

This large-signal transient gain of 1.997 V/V is in exact agreement with the 1.998 V/V target established during the closed-loop AC analysis.

6 Conclusion and Final Specifications

The fully-differential folded cascode OTA and its continuous-time transistor-level CMFB circuit were successfully designed, optimized, and verified in the GF180MCU 2.5 V process.

Table 5 compares the final simulated performance against the initial target specifications. The design successfully meets or exceeds all requirements, including the stringent 100 ns settling time and the 70° phase margin for both the differential and common-mode loops.

Table 6 provides the DC current budget and total power consumption. Table 7 provides an estimate of the active transistor area ($W \times L$) based on the extracted ADT dimensions.

Table 5: Final Performance Summary (Target vs. Verified)

Parameter	Target Specification	Simulated Result
Supply Voltage (V_{DD})	2.5 V	2.5 V
Load Capacitance (C_L)	500 fF	500 fF
Output Common-Mode Level	Nominal $V_{DD}/2$	927.75 mV (Swing Optimized)
Open-Loop DC Gain	≥ 60 dB	69.27 dB
Closed-Loop Gain (A_{cl})	2 V/V	1.998 V/V
Differential Phase Margin	$\geq 70^\circ$	89.15°
Common-Mode Phase Margin	$\geq 70^\circ$	70.72°
1% Settling Time	≤ 100 ns	98.09 ns
Differential Output Swing	1.2 V _{p-p}	1.198 V _{p-p} (Unclipped)

Table 6: Current and Power Consumption Summary

Sub-Circuit	Current / Power
OTA Core Current (Tail, Cascodes, Bias)	$\approx 59.4 \mu\text{A}$
CMFB Circuit Current (Buffers, Error Amp)	$\approx 29.1 \mu\text{A}$
Total System Supply Current	$\approx 88.5 \mu\text{A}$
Total Power Consumption ($V_{DD} = 2.5$ V)	$\approx 221.3 \mu\text{W}$

Table 7: Estimated Active Transistor Area

Block	PMOS Area (μm^2)	NMOS Area (μm^2)	Total Area (μm^2)
Main Folded Cascode OTA	100.6	18.1	118.7
Transistor-Level CMFB	181.2	19.0	200.2
Total Active Device Area*	281.8	37.1	318.9